21ASC305T
ROCKET PROPULSION

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- ***Unit-1 - Basic Concepts 9 Hour***
- *Rocket propulsion systems -working principle, Classification of rockets, Rocket equation, Rocket Nozzles and its classifications -Nozzle performance- equilibrium and frozen flow, Characteristic velocity and Thrust coefficient, Performance parameters and efficiencies of rocket, Staging and clustering of rocket*
- ***Unit-2 - Chemical Propellants 9 Hour***
- *Chemical propellants -Molecular mass, Specific heat capacities, Specific heat ratio, mixture ratio, Stoichiometric ratio and equivalence ratio, Energy release during combustion-Heat of formation and Heat of combustion-Combustion Instabilities-Criterion for choice of propellant, Solid Propellants-Composition and processing, Liquid Propellants-Classifications of liquid propellant, Storability of liquid propellant, Chemical Equilibrium*
- ***Unit-3 - Solid Rocket Engine 9 Hour***
- *Propulsion Elements for Solid Rocket Motors, Solid propellant combustion and Internal Ballistics of Motors -Mechanism of burning, Ignition System-Igniter types - Solid propellant grain design - Burn Rate-Factors influencing burn rates, burn rate index for stable operation, Action time and burn time, Design of Solid Propellant rocket, Simulation of solid rocket motor performance.*
- ***Unit-4 - Liquid Rocket Engine 9 Hour***
- *Liquid Propellant Rocket-Hardware components and its functions, Propellant feed systems –Gas pressure feed system, Turbo pump feed systems and engine cycles, Tank Pressurization - Types of injectors – Thrust chamber, Cooling of Thrust chamber - Cryogenic propulsion system and its special features, Simulation of Liquid rocket performance, Hybrid rockets- Review of solid-fuel regression rate behavior in classical and non-classical Hybrid Rocket Motors - Rocket engine design using software.*
- ***Unit-5 - Non-conventional Propulsion Techniques 9 Hour***
- *Non - conventional propulsion techniques -Electrical Rockets-Electro-thermal propulsion system-Arc-jet thruster, Resistojet Thruster-Electro-static thruster-Ion thruster, Bombardment Ionization, Hall Effect Thruster -Electro-magnetic propulsion system-Magneto plasma dynamic accelerator, Pulsed Plasma accelerator- Laser Propulsion -Nuclear powered rockets- Solar Propulsion system*

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- *1. George P. Sutton, Oscar Biblarz, "Rocket propulsion elements", Wiley India Pvt Ltd. Ninth Edition, 2017*
- *2. Ramamurthi.K, "Rocket propulsion", Laxmi Publications, India, Second edition, 2019.*
- *3. J D. Mattingly and K M. Boyer,"Elements of Propulsion: Gas Turbines and Rockets", Second Edition,2016*
- *4. Philip Hill and Carl Peterson, "Mechanics and thermodynamics of propulsion", Pearson India, second edition, 2010.*
- *5. Stephen R. Turns, "An Introduction to Combustion: Concepts", McGraw-Hill Education; 4th edition, 2020*
- *6. Yahya S M, "Gas Tables",8th Edition, 2018*

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Unit-3 - Solid Rocket Engine

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IGNITION PROCESS

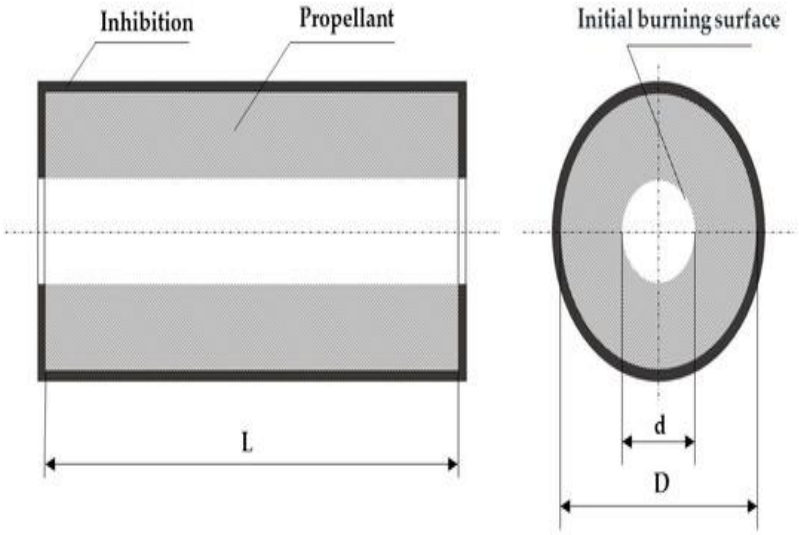
- Propellant ignitability is affected by several factors, including
- (1) propellant formulation,
- (2) propellant grain surface initial temperature,
  - A higher initial temperature means less external energy is required to heat the surface to its ignition point.
  - reduces the amount of external thermal energy required to bring the surface to its auto-ignition point, making it easier to ignite



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IGNITION PROCESS

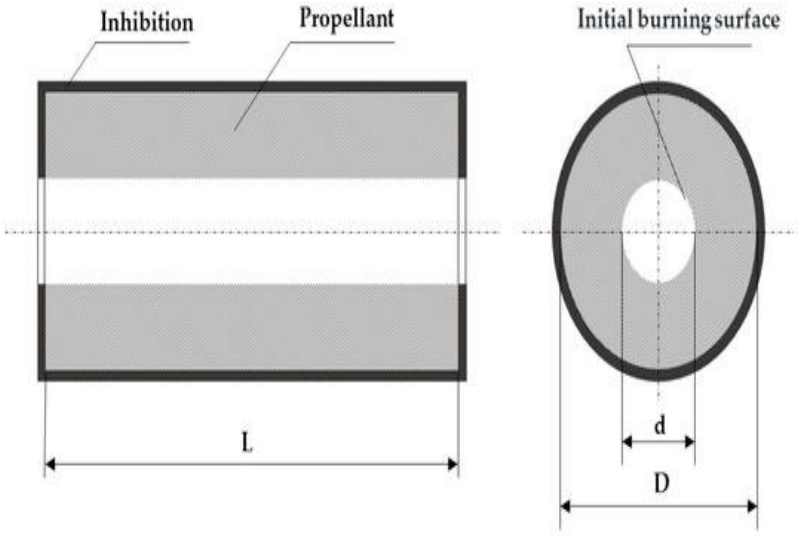
- (3) surrounding pressure,
  - Higher ambient or initial pressure increases the gas density, accelerates gas-phase chemical reactions, and enhances heat
- (4) modes of heat transfer,



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IGNITION PROCESS

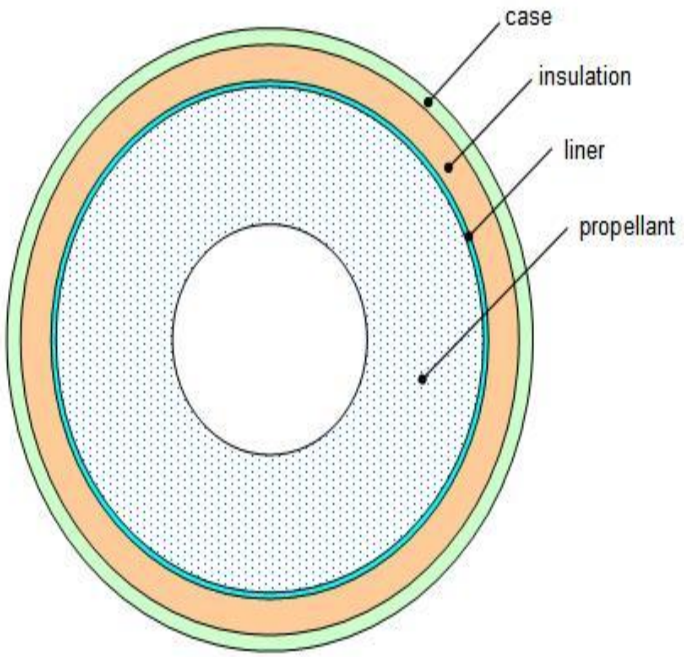
- (5) grain surface roughness,
  - Increased roughness creates microscopic cavities and a higher surface area, which can trap heat, create local hot spots, and enhance convective heat transfer.



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IGNITION PROCESS

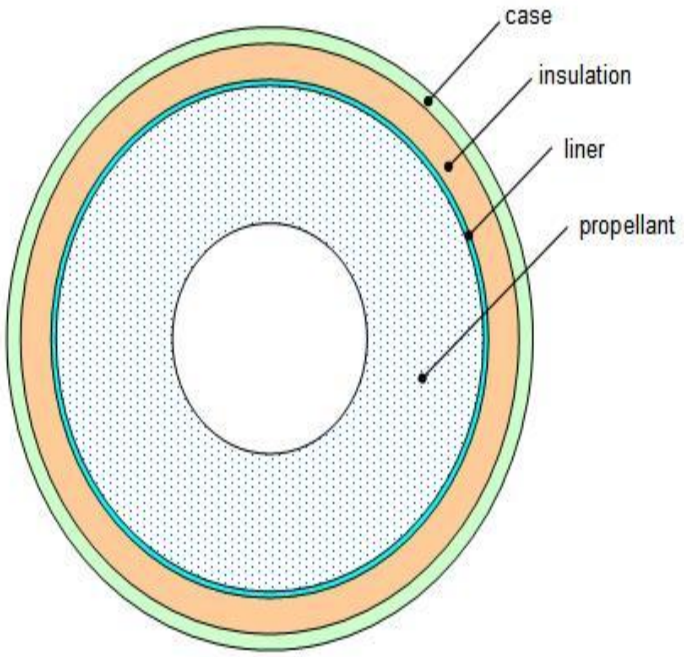
- Propellant ignitability is affected by several factors, including
- (6) propellant age,
  - Physical aging can also cause mechanical changes, such as binder-oxidizer dewetting or microscopic cracking, which drastically changes the burning surface area.



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IGNITION PROCESS

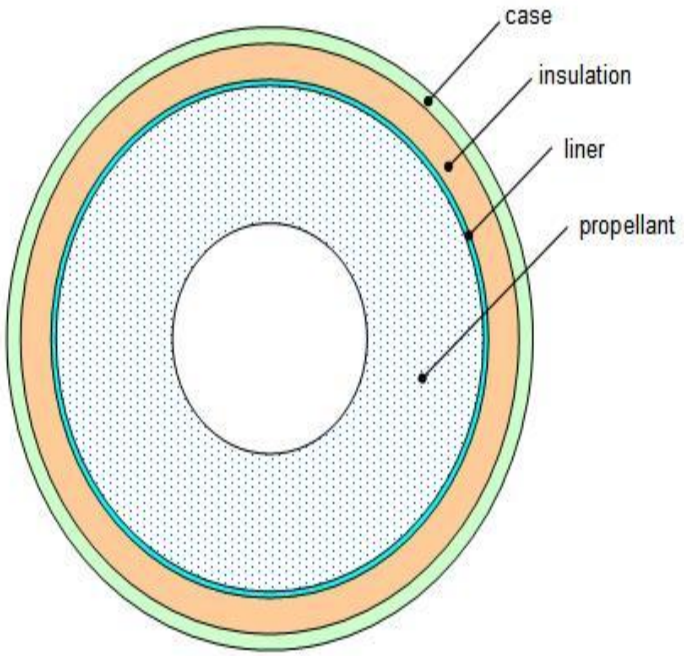
- (7) composition and hot-solid-particle content of igniter gases,
  - Igniter gases containing high concentrations of condensed-phase hot solid particles (like zirconium or boron oxides) provide highly localized, intense conductive heat transfer upon striking the grain surface, which is far more efficient than convective heating alone



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IGNITION PROCESS

- (8) igniter propellant and its initial temperature,
  - a colder igniter propellant burns slower and produces a lower mass flow rate, directly delaying the overall rocket motor ignition.
- (9) velocity of hot igniter gases relative to the grain surface
  - High gas velocity enhances the convective heat transfer coefficient, warming the grain faster. However, if the velocity is excessively high, it can cause a local cooling effect (erosive cooling)
- (10) cavity volume and configuration.



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Inadvertent Ignition

- If a rocket motor is **unexpectedly ignited** and starts combustion, the very hot exhaust gases may cause burns and local fires, or ignition of adjacent rocket motors.

- **Unless the motor is constrained or fastened down**, its thrust will accelerate it to unanticipated high velocities or erratic flight paths that can cause much severe damage.

- Its exhaust cloud can be toxic and corrosive.

- Inadvertent ignitions may be caused by the following effects:

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Inadvertent Ignition

- **Stray or induced currents** that activate the igniter.
- **Electrostatic charging** causing unintended sparks or arc discharges.
- **Fires excessively heating rocket motor exteriors**, raising the solid propellant temperature above its ignition point.

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Inadvertent Ignition

- **Impacts** (bullet penetration or dropping the rocket motor onto a hard surface).

- **Energy absorption from prolonged mechanical vibrations** that overheat the propellant (e.g., transport over rough roads).

- Radiation from nuclear explosions.

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IGNITER HARDWARE

- Since **igniter propellant mass is small** (often less than 1% of the motor propellant) and burns mostly at low chamber pressure (hence low I_s), it has a negligible contribution to the motor's overall total impulse.

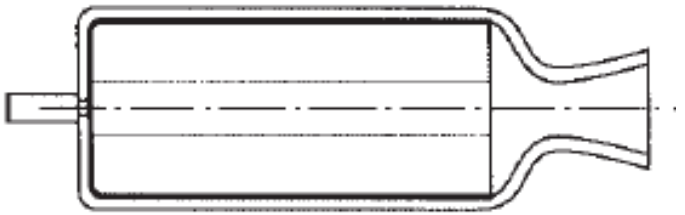
- When mounted at the forward end, the gas flow over the propellant surface helps to achieve ignition.

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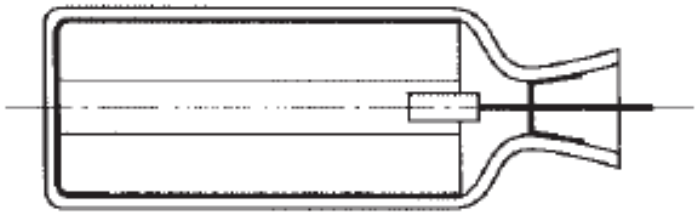
Mounting options for igniters



Aft, internal



Aft, external



Forward, internal
(supported by nozzle exit cone)



Forward, external

Mounting options for igniters

- **Forward, Internal** : The igniter is mounted entirely inside the motor case at the closed front dome. It shoots a stream of fire directly down the central cavity toward the nozzle. This is the most common setup for large space boosters.

- **Forward, External** : The main body of the igniter is attached to the outside of the front dome casing. Only its nozzle or port protrudes inside. This keeps the igniter hardware separate from the high-pressure combustion chamber.

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Mounting options for igniters

- **Aft, Internal** : The igniter sits at the rear near the nozzle throat. It is held in place by a support rod attached to the nozzle exit cone. Once the main motor ignites and builds pressure, the entire support and igniter assembly is intentionally **blown out the back of the nozzle** to clear the exhaust path.

- **Aft, External** : The igniter is mounted on the outside of the rear dome or nozzle housing, shooting its ignition gases upward into the propellant cavity from the back.

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AMICS

A solid rocket motor operates at a steady-state chamber pressure of $P_c = 6.5\ \text{MPa}$. The composite propellant has an empirical burn rate constant $a = 4.8\ \text{mm/s} \cdot (\text{MPa})^{-n}$ and a pressure exponent $n = 0.38$.

1. Calculate the linear burning rate (r) of the propellant at this operating pressure.

2. If the propellant grain is a end-burning cylinder with an initial length of **450 mm**, determine the total burn time (t_b) of the motor.

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(Deemed to be University u/s 3 of UGC Act, 1956)

$$r = a \cdot P_c^n$$

$$r = 4.8 \cdot (6.5)^{0.38}$$

$$(6.5)^{0.38} \approx 2.0327$$

$$r = 4.8 \cdot 2.0327 = \mathbf{9.76}\ \text{mm/s}$$

$$t_b = \frac{L}{r}$$

$$t_b = \frac{450\ \text{mm}}{9.757\ \text{mm/s}} = \mathbf{46.12}\ \text{seconds}$$

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A small solid rocket motor generates a steady gas mass flow rate of $\dot{m} = 2.5$ **kg/s**. The design chamber pressure is $P_c = 4.0$ **MPa** (4.0×10^6 Pa) and the propellant exhaust gas has a characteristic velocity $c^* = 1,300$ **m/s**. Calculate the required **nozzle throat area** (A_t) in m^2 to maintain this operating condition.

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$$A_t = \frac{\dot{m} \cdot c^*}{P_c}$$

$\dot{m} = 2.5$ kg/s

$c^* = 1,300$ m/s

$P_c = 4.0 \times 10^6$ Pa

$$A_t = \frac{2.5 \cdot 1,300}{4.0 \times 10^6}$$

$$A_t = \frac{3,250}{4,000,000} = \mathbf{0.0008125}\ \text{m}^2\ (\text{or } 812.5\ \text{mm}^2)$$

IGNITER HARDWARE

- The two basic most common types,
- *pyrotechnic igniters* and
- *pyrogen igniters*

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IGNITER HARDWARE

- In industrial practice, pyrotechnic igniters are defined as igniters that use solid explosives or energetic propellant-like chemical formulations (usually **small pellets** of propellant that provide large burning surfaces and short burning times) as the heat-producing materials.

- produces a **large amount of hot gas/flame for a short time** to ignite the main solid propellant.

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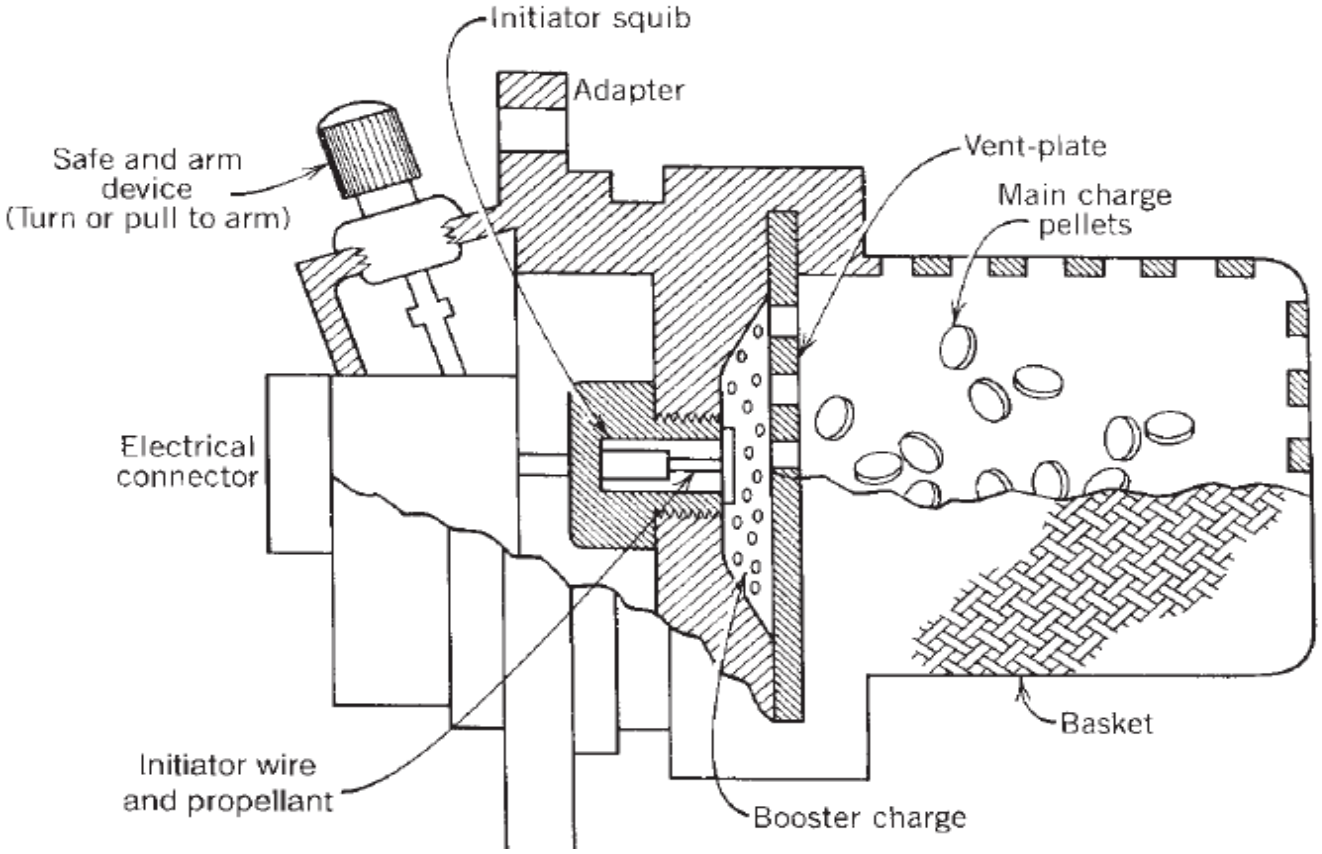
IGNITER HARDWARE

- This definition fits a wide variety of designs, known in the trade as
  - bag and carbon igniters,
  - powder can,
  - plastic case,
  - pellet basket,
  - perforated tube,
  - combustible case,
  - jellyroll,
  - string, or
  - sheet igniters.

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Type	
Bag igniter	Energetic material contained in a bag
Carbon igniter	Uses a carbon-based container/structure
Powder-can igniter	Material contained in a can-like enclosure
Plastic-case igniter	Material enclosed in a plastic case
Pellet-basket igniter	Pellets held in a basket-like structure
Perforated-tube igniter	Material contained in a tube with holes
Combustible-case igniter	Case itself is designed to burn
Jellyroll igniter	Material arranged in a rolled configuration
String igniter	Energetic material arranged as a string
Sheet igniter	Material arranged in sheet form

Typical pyrotechnic igniter with three different propellant charges that ignite in sequence.



- **Ignition sequence**
- **Electrical signal → Squib/primer → Booster → Main ignition charge → Main propellant**
- **1. Electrical signal**
- When an electrical signal is applied to the **initiator**, it activates a small sensitive pyrotechnic material.
- **2. Squib / primer charge**
- The initiator contains a small amount of **sensitive powdered pyrotechnic material**, called the:
  - **Squib**, or
  - **Primer charge**
- It produces a small but intense amount of **heat and hot gases**.

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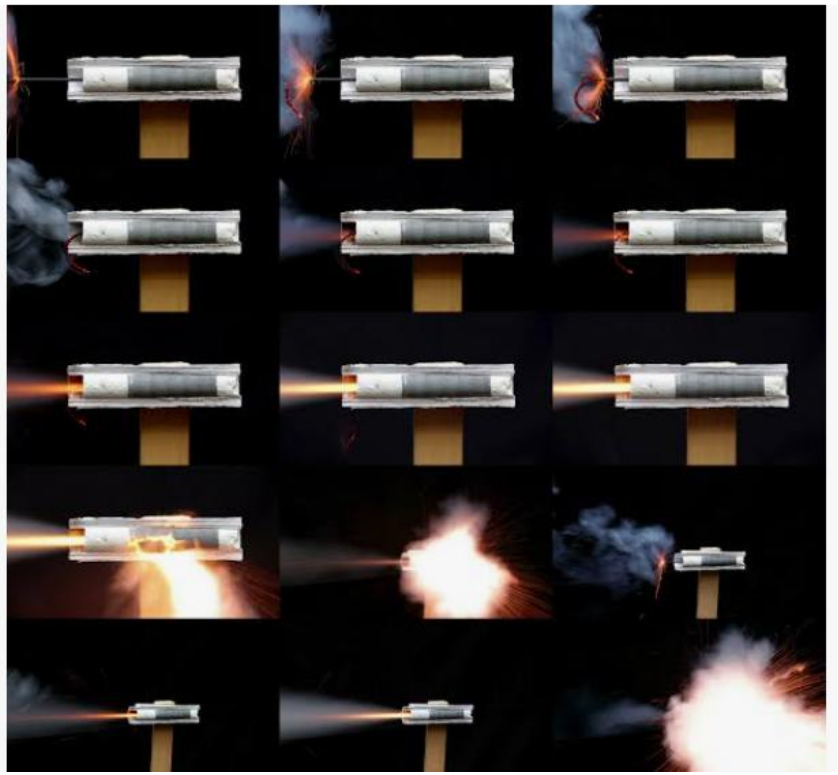
- **3. Booster charge**
- The heat from the squib **ignites the booster charge**.
- The booster contains a larger quantity of energetic material than the squib, so it produces substantially more heat and hot gases.
- **4. Main ignition charge**
- The booster then provides sufficient heat and hot gases to ignite the **main ignition charge**.

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Pyrotechnic Igniters

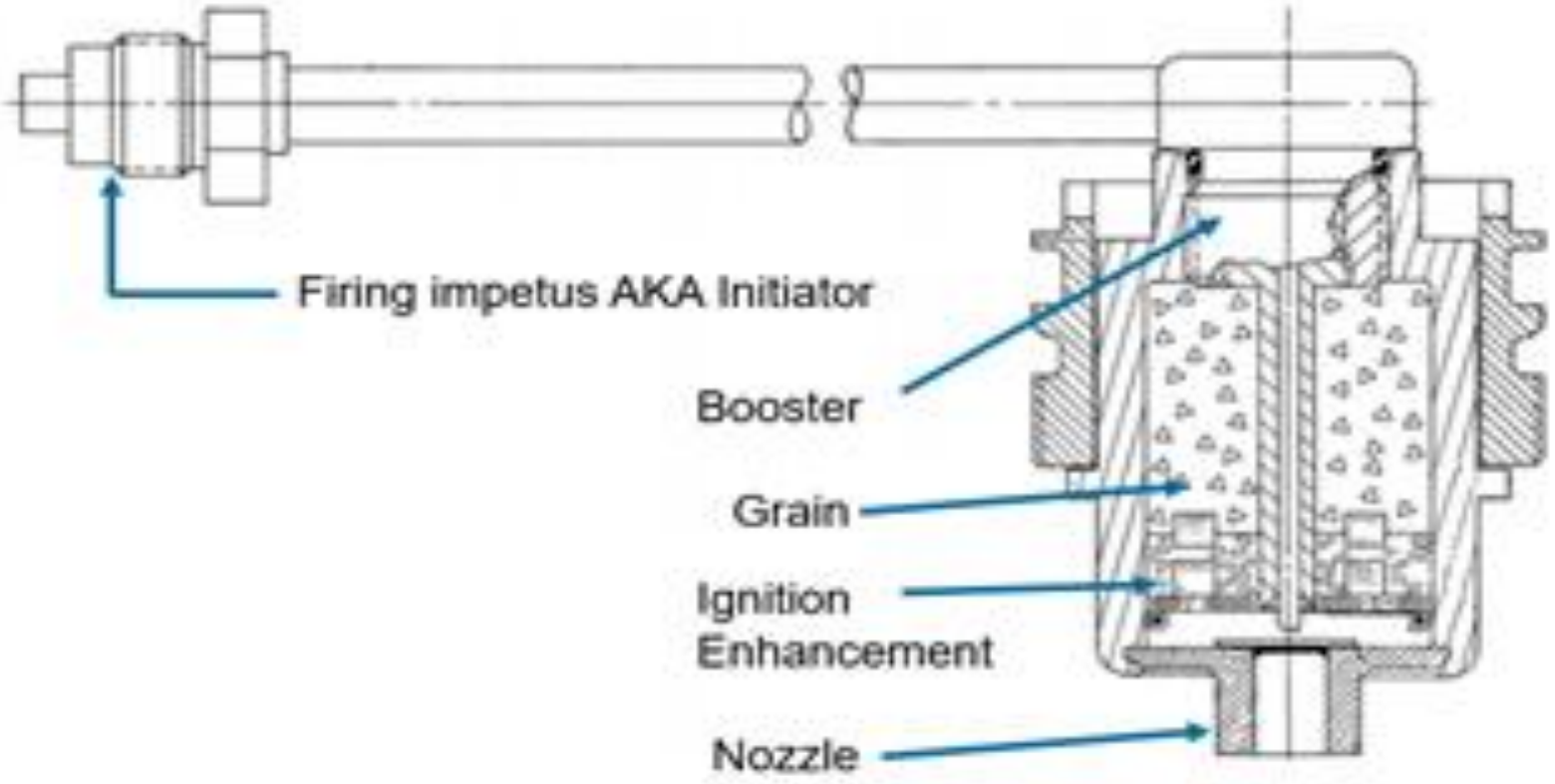
- A special form of pyrotechnic igniter is the *surface-bonded* or *grain-mounted igniter*.

- **Surface-bonded or grain-mounted igniters** are pyrotechnic components applied directly to the surface of a solid rocket propellant grain.



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Pyrotechnic Igniters



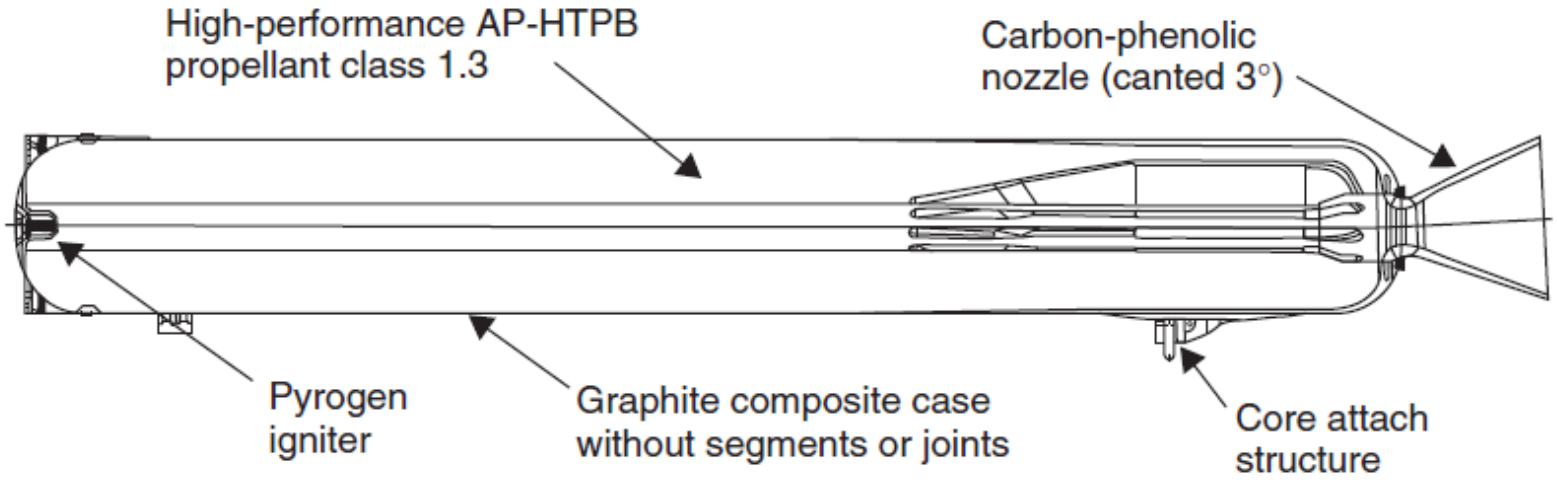
Pyrogen Igniters

- A pyrogen igniter is basically a small unit, containing all the elements of a rocket motor, used to ignite a larger rocket motor.

- They all consist of one or more nozzle orifices (both sonic and supersonic types) and most use conventional rocket motor grain formulations (sometimes the same as the main propellant grain) and design technology.

- Heat transfer from the **pyrogen** gas to the motor grain is largely convective, with hot gases contacting the inner grain surface, in contrast to **pyrotechnic igniters** that transfer heat by high-energy radiation.

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Thrust = 287,346 lbf (vacuum average at 73°F)
Thrust = 374,730 lbf (vacuum peak at 73°F)
Specific impulse = 279.3 sec (vacuum average)
Burn time = 89.43 sec (nominal)
Nozzle expansion ratio = 16.1 at ignition, 11:1 at burn-out
Chamber temperature = 6319°R (at 1000 psia)
Propellant mass = 93,656 lbm (nominal), expended inerts = 942 lbm
Length = 67 ft
Diameter = 62 in
Ambient temperature limits = 20 to 120°F
Propellant = AP(68%)/AI(19%)HTPB(13%)
Insulation = EPDM polymer (internal, erosion resistant)

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Pyrogen Igniters

- For pyrogen igniters, **initiator and booster charge designs** are often very similar to those used in pyrotechnic igniters.

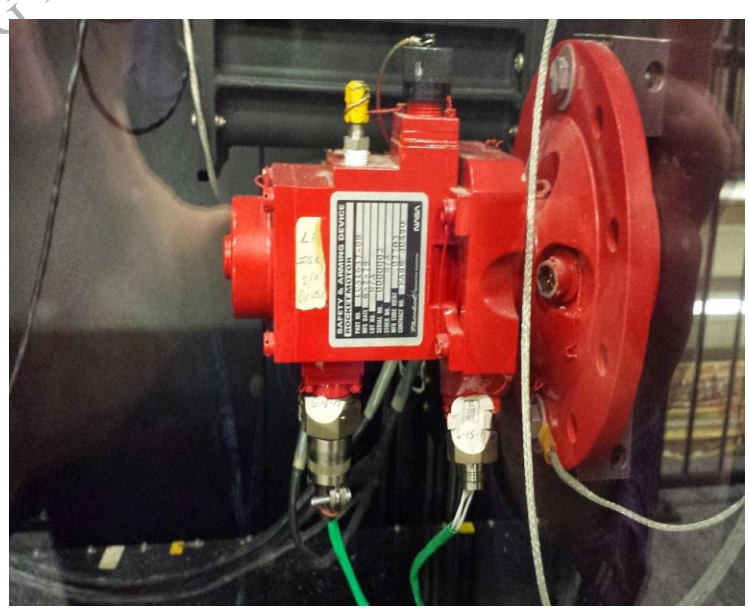
- Reaction products from the main charge impinge on the inside surface of the rocket motor grain, producing motor ignition.

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Pyrogen Igniters

- Two approaches are commonly used to safeguard against rocket motor misfires, or inadvertent ignition;
- one is to use of the classical ***safe and arm device*** and
- the second is to ***design safeguards*** into the initiator.

- ***Safe and arm device***



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- **SAFE position**
- The S&A device prevents ignition in **two possible ways**:
- **1. Electrical blocking**
- The electrical connection to the initiator is interrupted/grounded:

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- **2. Mechanical blocking**
- A barrier/rotor is positioned so that the ignition path is not aligned.

- As the device is moved into the ***arm*** **position**, the electric ignition circuit is no longer blocked and an ignition flame can reliably propagate from the igniter's booster and main charges to the propellant surface.

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Pyrogen Igniters

- Energy sources causing unintentional ignitions (usually disastrous when they happen) can be
-
- (1) static electricity,
- (2) induced electrical currents from electromagnetic radiation—such as radar,
- (3) induced electrical currents from ground test equipment, communication apparatus, or nearby electrical circuits in the flight vehicle, and
- (4) heat, vibration, or shock from motor handling and operations.

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Pyrogen Igniters

- **Electric initiators** in motor igniters are also called squibs, glow plugs, primers, and headers; they always constitute the initial element in the ignition train and, if properly designed, can act as a safeguard against unintended ignition.

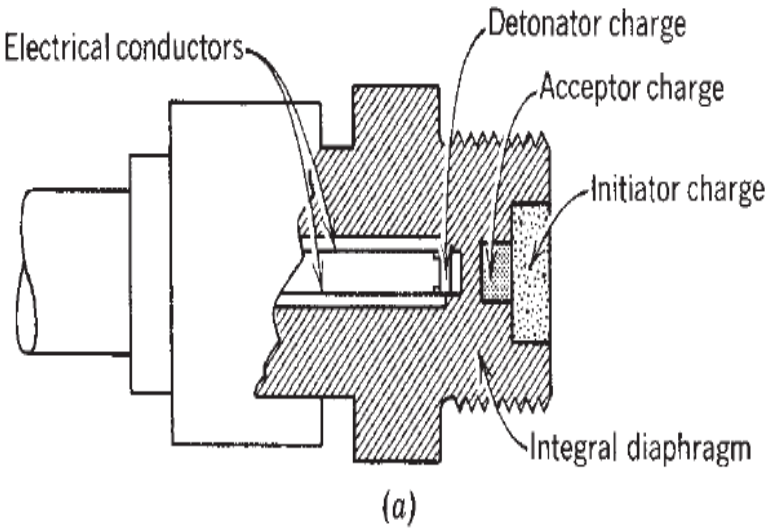
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Pyrogen Igniters

- In the integral diaphragm type (*a*), the initial ignition energy is passed in the form of a shock wave through the diaphragm activating the acceptor charge, with the diaphragm remaining integral.

- The same principle is also used to transmit shock waves through metal case walls or metal inserts in filament-wound cases; here, the combustion case would not need to be penetrated and can remain sealed.

- **Integral diaphragm type**

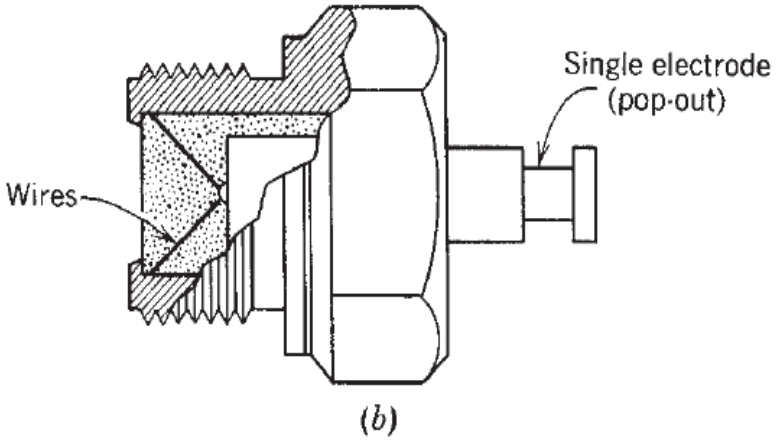


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Pyrogen Igniters

- The header in type (*b*) resembles a simple glow plug with two high-resistance bridgewires buried in the initiator propellant charge.

- **Header type with double bridgewire**

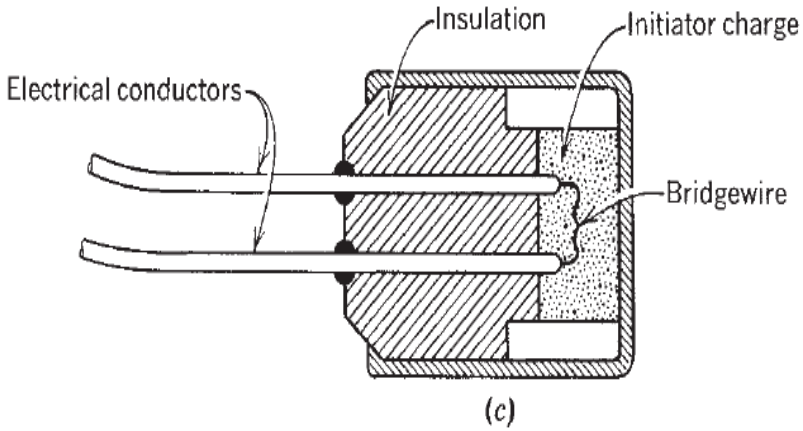


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Pyrogen Igniters

- The design in type (*c*) employs a small bridgewire (0.02 to 0.10 mm) of low-resistance material, usually platinum or gold, that explodes by the application of a high-voltage discharge.

- **Exploding bridgewire type.**



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Mass Flow Relations

- $\dot{m} = A_b r \rho_b$
- The **gaseous propellant mass** evolving from a burning surface per unit time must equal the sum of the change in gaseous mass storage per unit time in the combustion chamber (due to increases in volume of the grain cavity) and the mass flowing out through the exhaust nozzle per unit time,

$$A_b r \rho_b = d(\rho_1 V_1)/dt + A_t p_1 / c^*$$

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Mass Flow Relations

- first term on the right represents the rate of change in storage of the hot gas in the combustion chamber volume.
- last term represents the propellant flow rate through the nozzle
- ρ_1 is the chamber hot-gas density
- V_1 represents the rate of increase chamber gas cavity volume

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Mass Flow Relations

- Though the rate of change of hot gas in the grain cavity is always important during start-up, it is seldom included in preliminary designs.

- The magnitude of the burning surface A_b may or may not change significantly with time and this is a function of grain design.

- For preliminary performance calculations, the nozzle throat area A_t is usually taken as constant

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Mass Flow Relations

- for the entire burning duration but, for accurate performance predictions **during hot firings**, it is necessary to account for oxidation, erosion and abrasion in the nozzle material which increases the nozzle throat area as the propellant burns; such nozzle enlargements are usually small (0.05 to 5%).

- The chamber gas volume V_1 will measurably increase with burn time.

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Mass Flow Relations

- The commonly used relation for the pressure in steady burning
  - $p_1 = K\rho_b r C^*$ where $K \equiv A_b/A_t$
- For steady flow and steady burning, the value of K (or A_b) must remain essentially constant.

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Burning Rate Relation with Pressure

- With many propellants it is possible to approximate the *burning rate* as a function of *chamber pressure*, at least over a limited range of chamber pressures.

- For a majority of production-type propellants the most commonly used empirical equation is
  - $r = aP_1^n$

- when $n > 1.0$, any pressure disturbances present will be amplified in the chamber.

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Burning Rate Relation with Pressure

- For any particular propellant and for wide temperature and pressure limits, the burning rate can vary by factors of 3 or 4.
- For all propellants this translates to a range from about 0.05 to 75 mm/sec or 0.02 to 3 in./sec.
- To achieve such rates, combinations of very small sized ammonium perchlorate (AP), burning rate catalysts, additives, or embedded metal wire are needed.

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When the size of the AP particles is reduced or the percentage of AP is increased, an increase in burning rate is observed

